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The Completion

****A relativistic field theory carrying $a_0 = \kappa c \sqrt{(G\rho - \Lambda)}$ ****

Zimmerman, Carl P. — Briar Creek Tech Assembled 2026-08-09; **v2, v3, v4, v5 same day**. Every numbered property below is backed by a committed, runnable script that exits non-zero on failure. Scripts named at each line. **v5 changes — the largest revision, and it runs against the programme’s interest**. A six-stage sequence (nbody_2026/, every stage committed and green) settles the galaxy-interior question that v3 reopened as non-claim 2d, and **settles it adversely: the Q-sector cannot also be absent from galaxies, and the repair space of local, energy-preserving modifications is provably empty**. Three results carry it: (i) *the dust mass IS the conserved shift charge* ($\rho = Q_0 n$ with $\rho/n = Q_0$ regardless of any added Y-dependent Q-mass), so it cannot be suppressed locally — which retro-explains three withdrawn mechanisms; (ii) *unlocking the charge does not unlock the energy* ($\nabla_\mu T^{\mu\nu} = 0$ holds regardless, and transport out of a galaxy takes 690 Gyr); (iii) **** dynamics responds to $\rho + 3p$ but lensing responds to $\rho + p$, and no equation of state kills both**** — so the same no-slip identity that underwrites row 1 forbids hiding the dust. Also corrected here: v3/v4’s claim that the DBI cap bounds the dust *density* (it bounds the pressure; *ρ_{exc} is linear in u and diverges at saturation*), and v4’s *spliced amplitude band*. **Rows 1–12 and 15–17 are untouched. v2 changes:** row 13’s *shift-charge cluster route* is **withdrawn** (killed by the 1-Mpc confrontation: *smooth accretion restores $\$ \rightarrow 1$ for any cold IC*) and replaced by the *** $a\{0\}$ –bump environment response ***, which passed a five–environment matrix AND its perturbation health check; the health check carved two hard constraints $\leq 8.4 \times 10^{-7}$; amplitude $\leq 2.7 \times$ fiducial). The coefficient row now reads as **measured:** $\kappa = 0.551 \pm 0.043, \pm 0.063$ if the H_0 tension is carried as a systematic ($\kappa \propto a_0/H_0$). **v3 changes:** (i) the **full AeST perturbation matrix** for the bump term is now DONE and added as row 17 — $c_T = 1$ exact, no-ghost as a theorem, quasi-static closure derived, amplitude cap softened to a band; non-claim 2c is updated accordingly. (ii) **Non-claim 2 is corrected:** v2’s “by item 13 it is absent where rotation curves are measured” was a dangling pointer — row 13 was withdrawn *in the same version* — so the galaxy-interior status of the Q-sector dust is **REOPENED** and now stated as its own open problem (non-claim 2d). v2 carried this internal inconsistency for several hours; the correction runs against the framework’s interest and is flagged rather than hidden. (iii) §3’s amplitude cap now states its units ($4.5 \text{ Mpc}^{-2} = 2.7 \times$ the fiducial 1.65 Mpc^{-2}).

1. The action

$$S = \int d^4x \sqrt{-g} \left[\underbrace{\frac{R - 2\Lambda_{\text{bare}}}{16\pi G}}_{\text{Einstein-Hilbert}} + \underbrace{\mathcal{L}_{\text{aether}}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu = -1} + \underbrace{\mathcal{F}(Y, Q)}_{\text{the dark sector}} \right] + S_{\text{matter}}[g, \psi]$$

with the two scalar invariants built from the khronon φ and the aether A^μ ,

$$Q \equiv A^\mu \nabla_\mu \varphi, \quad Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi.$$

This is **Aether-Scalar-Tensor** (Skordis & Złośnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework’s content is entirely in the choice of the free function , which **factorises**:

$$\boxed{\mathcal{F}(Y, Q) = \underbrace{\frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} + \underbrace{K(Q)}_{\text{dark energy + dark matter}} + \underbrace{A B\left(\frac{Y}{a_0^2}\right) (Q - Q_0)^2}_{\text{a}_0\text{-bump response (v2)}}$$

$$B(y) = \frac{y}{(1+y)^2}$$

The third term (v2) is a **position-dependent Helmholtz mass** $\mu_{\text{eff}}^2 = A B(g^2/a_0^2)$ **peaked at the framework’s own** a_0 — the location costs nothing (Y/a_0^2 is already the Y-sector’s normalisation); one new calibrated amplitude $A \approx 1.7 \text{ Mpc}^{-2}$. *The dark sector’s response is resonant at the MOND transition, and clusters are the cosmic objects that live at a_0 (R_{500} at 0.33–0.58 a_0).*

1.1 The Y-sector — MOND

$_Y$ is fixed by the framework’s Route A kernel $\nu(y) = 1/(1 - e^{\wedge}(-\sqrt{y}))$, $y = g_bar/a_0$. In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \quad x(u) = \frac{u^2}{\mu(u)}, \quad u = \sqrt{y}, \quad x = g_{\text{obs}}/a_0.$$

with the normalisation

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = \frac{cH_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad \kappa = \frac{1}{2}, \quad Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$ is **FITTED, not derived**. See §4.

1.2 The Q-sector — an offset DBI khronon

$$K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}} \right], \quad u \equiv Q - Q_0, \quad |u| < \Lambda_D$$

Three jobs from one function:

piece	limit	what it is
the offset $-M^4$	$u = 0, K = 0$	$\rho = -K = M^4, p = K = -M^4$ $\Rightarrow \mathbf{w} = -1$ exactly : dark energy
the deviations	$u \ll \Lambda_D$	$K \rightarrow \mu^2 u^2 / 2$ is the pressure , while $\rho_{\text{exc}} = Q_0 \mu^2 u$ is linear in $u \Rightarrow w = u / 2Q_0 \rightarrow 0$: dust , i.e. the dark matter. (<i>v5: the linearity is why the dust mass equals the conserved charge — see non-claim 2d.</i>)
the saturation	$u \rightarrow \Lambda_D$	K bounded, $K \rightarrow \infty \Rightarrow w \rightarrow 0$ again: kills the early stiff phase

Parameters: $\mathbf{M}$ (vacuum scale, $M^4 = \rho_{\Lambda}$), μ (Helmholtz mass), $**\Lambda_D**$ (DBI field-space scale), $\mathbf{I}_0$ (the conserved shift charge = the dust amount).

2. What is verified

#	property	value / statement	script
1	Lensing — the test that killed modified inertia	$\Phi = \Psi \Rightarrow \gamma_{\text{PPN}} = 1, M_{\text{dyn}}/M_{\text{lens}} = 1$ exactly. $21.2\sigma \rightarrow 0.601\sigma$	mi_relativistic_completion_aest.
2	Kernel embeds in $_{\text{Y}}$	deep-MOND $\mu \rightarrow x$ exact; Newtonian $\$ \rightarrow 1; x(u) \text{bijection}(h = e^{(-u)}(1+u) > 0); \text{free function} * \text{convex} * * \text{same} 3 * \text{Newtonian residual} * * e^{(-\sqrt{\$}y)} = 3.6 \times 10^{-3457}$ at Earth's orbit	same

#	property	value / statement	script
4	Dark energy is not an input	$w = -1$ exactly at the condensate minimum — proved, not assumed	mi_condensate_vacuum_energy_a0_2
5	Ghost-free over the whole field range	$K = \mu^2(1-s^2)^{-3/2} > 0$ for all $ s < 1$	mi_dbi_khronon_2026.py
6	Subluminal everywhere	$c_s^2 = \Lambda_D$ $s(1-s^2)/(1+\Lambda_D s) \leq 0.385 \Lambda_D$	same
7	The published no-go is dissolved	Blanchet & Skordis 2024 §4.3.1's 455× conflict reverses sign : bounded pressure $\Rightarrow w \rightarrow 0$ early $\Rightarrow$ cosmology becomes a <i>lower</i> bound on μ^{-1} , same direction as MOND	same
8	Theorem : why no polynomial works	for $K \sim u^n$, early-time $w \rightarrow 1/(n-1)$. Quadratic 1, quartic $\frac{1}{3}$, sextic — all fail. Only boundedness gives $w \rightarrow 0$	same
9	CMB acoustic peaks — real CLASS run	at $\Lambda_D \leq 10^{-2}$, $c_s^2(\text{recomb}) = 2.9 \times 10^{-8}$; TT unchanged to 0.069% , $P(k=0.2)$ to 1.71%. Indistinguishable from CDM	mi_dbi_cmb_class_run_2026.py
10	BTFR	exact, from convexity	mi_route_a_field_theory_2026.py
11	g^{-2} Lorentz violation	restored by the aether (pure Bekenstein–Milgrom had lost it)	mi_relativistic_completion_aest
12	a_0 tied to Λ	$a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ with $m_{\text{cond}} = M^2/(\sqrt{2}M_{Pl}), M^{\{4\}} = \rho \Lambda$ — agrees to 0.076%	mi_condensate_vacuum_energy_a0_2

#	property	value / statement	script
13	Clusters (v2)	shift-charge IC route withdrawn: killed by the 1-Mpc confrontation — cluster R500 and galaxy-outskirt lensing share a scale with disjoint requirements, correlated Gaussian ICs cannot select environments, and smooth accretion restores \$ → 1 for any cold IC * *. Replaced by the * * a {0} — bump response * *(row15) mi_c,oute_1 mpa auto fiducial n_2026.py (9/9) 14 * * Directional EF E signal * * pure MI predicted * exactly zero; AQUAL — class predicts 1–4 15 * * a {0} \$-bump environment response	passes all five environments: cluster (calibration, $\mu^2_{\text{eff}} = 0.23 \text{ Mpc}^{-2}$), galaxy interior 0.034% of M_b (RAR cost 4×10^{-4} dex), galaxy 1 Mpc 1.2% vs the 5.9% strict bound, linear cosmos 0.6% of mean matter, solar 10^{-19} . Vanishes on FRW (Y=0) and is second-order in perturbations $\Rightarrow$ linear CMB/P(k) construction. Evades both prior kills (a response has no charge to advect; couples to environment directly)
16	...and its health check	no ghosts ever (kinetic contribution $2AB \geq 0$, $= 0$ on FRW), no Ostrogradsky sector; FRW gradient health **excludes the old $\Lambda_D = 10^{-2}$ and forces $\Lambda_D \leq 8.4 \times 10^{-7}$ (window stays 3.7 orders); halo gradient health caps $A \leq 2.7 \times$ fiducial, excluding the demanding end of Mistele's cluster band — a two-sided falsifiable pinch; free signature: cluster anisotropic stress at $O(0.6)$	mi_a0_bump_health_2026.py (11/11)

#	property	value / statement	script
17	...and the FULL perturbation matrix (v3)	with all aether mixings: c_T = 1 exact (the new terms are algebraic in the metric — GW170817 safe by construction); no-ghost is a THEOREM (the bump’s kinetic addition is rank-1 PSD in the $\chi\chi$ entry $\implies$ Weyl monotonicity: it cannot lower any eigenvalue of AeST’s healthy kinetic matrix; closed-form $2\times 2 +$ 200 random 4×4); unit-norm gives $\delta A^0 =$ $-\Phi A^0 \implies \delta Q = \chi -$ $Q_0\Phi$ — the quasi-static phenomenology of row 15 is derived, not assumed ; gradient block exact: $\Delta G =$ $2\lambda[[q,$ $2y^{\{3/2\}}B],[2y^{\{3/2\}}B,$ $yq]], q =$ $(3y^2-8y+1)/(1+y)^4$; integrating out the aether softens the amplitude cap: at $\text{base_a} = 1$ the band is A_max $\in [2.72,$ 4.46] $\times$ fiducial $=$ [4.49, 7.36] Mpc^{-2} (v4 correction : the earlier “(2.7–7.4) $\times$ fiducial” spliced fiducial units with Mpc^{-2} , making the band look $1.66\times$ wider at the top than it is — base_a_lookup_and_robustness_2026.py, 12/12). The base_a lookup is now DONE and it 6 converts the caveat rather than closing it : base_a is built from AeST’s single	mi_aest_full_matrix_bump_2026.py (11/11)

3. Parameter count, honestly

	Λ CDM	this completion
dark sector	Ω_c, Ω_Λ	$M (= \rho_- \Lambda^{\{1/4\}}), I_0, \mu, \Lambda_D, A$ (v2)
MOND scale	—	a_0 — not independent , = $m_{\text{cond}}/(4\sqrt{\pi})$ via item 12
bounds	—	$\Lambda_D: 1.9 \times 10^{-10} \ll \Lambda_D \leq 8.4 \times 10^{-7}$ (health-bounded, v2 ; 3.7 orders); $A \approx 1.7$ Mpc^{-2} calibrated on clusters, health cap $A \leq 4.5 \text{ Mpc}^{-2} = 2.7 \times \text{fiducial}$ (row 16; the full matrix widens it to $[2.72, 4.46] \times \text{fiducial} = [4.49, 7.36] \text{ Mpc}^{-2}$ at $\text{base}_a = 1$ — row 17, where the base_a fork and its robustness are stated); $\mu^{-1} = 4392 \text{ Mpc}$ (item 12); $I_0 \approx \Omega_{\text{dm}}$ (an IC)

So: **five dark-sector numbers against Λ CDM's two** (v2 adds A), with a_0 derived from one of them rather than added. Not fewer parameters. What it buys is that Λ , dark matter and MOND come from **one function** instead of three unrelated sectors.

4. What is NOT claimed — read this before quoting anything above

1. $\kappa = \frac{1}{2}$ **IS NOT DERIVED**. Item 12 looks like a derivation and is not. $a = m_{\text{cond}}/(4\sqrt{\pi})$ is *algebraically identical* to $a = \sqrt[4]{G \Lambda}$ and to $Z = \sqrt[4]{(8/3)}$ — three costumes, one statement — and the residue is **convention-dependent** ($4\sqrt{\pi}$ with the reduced Planck mass, $\sqrt[4]{2}$ with the non-reduced). A derived number cannot depend on a bookkeeping choice. It is a **relabelling**, as this corpus's own theorem already established. What genuinely moved: “why $\kappa = \frac{1}{2}$?” becomes “why is $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$?”, which is sharper because m_{cond} is a *derived* quantity of a real field theory rather than a definition.
2. **Dark matter EXISTS**, at the full Ω_{dm} . Removing the pressureless component moves H_3/H_1 by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). What does **not** exist is a dark-matter *particle*: it is the Q-sector of the same scalar that supplies Λ and MOND. (**v3 correction**: v2 continued “and by item 13 it is absent where rotation curves are measured” — a dangling pointer, since v2 itself withdrew row 13. See 2d. **The defensible slogan shrinks to “no dark-matter particle” — never “no dark matter”; and (v5) “none in galaxies” is WITHDRAWN as a claim of this theory, not merely open — see 2d.)** 2d. (**v5**) **THE**

Q-SECTOR CANNOT ALSO BE ABSENT FROM GALAXIES — FALSIFIED WITHIN THE THEORY AS WRITTEN, AND THE REPAIR SPACE IS NOW PROVABLY EMPTY. This is the largest change since v1 and it runs against the programme’s interest. v2 asserted “none in galaxies” via a pointer to withdrawn row 13; a six-stage sequence (nbody_2026/, all scripts committed, 20/20 + 15/15 + 12/12 + 15/15

- 15/15 + 11/11) then settled the question the pointer had concealed.

The physics. The 1-Mpc smooth-accretion theorem implies the Q-sector dust falls into *every* collapsing basin, galaxies included. What happens next was the open problem, and it closed in four steps:

- **Nothing stops the collapse.** Wave (k^4 ghost-condensate) pressure gives a soliton scale of **0.18 AU** at halo density — eight orders below the RAR region — and it *shrinks* as ρ grows ($\lambda \propto \rho^{-\frac{1}{4}}$); a 1 kpc wave core would need M below the Lyman- α fuzzy-DM floor. The dust *is* an $n = 1$ polytrope ($p = K\rho^2$, $c_s^2 = 2K\rho$ rising) with a genuine **mass-independent 105 pc** equilibrium — but that lies **14.4× past DBI saturation**, where $c_s \rightarrow 0$ and the stiffening switches off. Endpoint: a black hole of the captured share, **falsified** $5.8 \times 10^5 \times$ **against Sgr A*** (4.3×10^6 M from individual stellar orbits — no mass model, kernel or cosmology in the comparison). *(v5 corrects v3/v4 here: an earlier draft claimed the DBI cap bounds the density and yields a ~ 250 pc core. $\mu^2 u^2/2$ is the **pressure**; exactly, $\rho_{\text{exc}} = Q_0 \mu^2 u$ is **linear** in u and diverges at saturation. Withdrawn.)*
- **THEOREM: the dust mass IS the conserved shift charge.** $\rho = Q_0 \cdot n$ with n the shift-charge density, and $\rho/n = Q_0$ **independently of any Y-dependent Q-mass, of any shape or amplitude.** So no such term can change how much dust a galaxy carries — *a conserved charge cannot be suppressed locally, only moved (gravity moves it inward) or not conserved.* This single statement retro-explains three withdrawn mechanisms: the shift-charge IC route, the Φ^n response, and an acceleration-gated suppressor built and killed in stage 4–5 (its eight gates all tested the field amplitude u ; the observable is n , and at fixed charge the suppressor makes a region a *cheaper* place to store charge — it attracts what it was built to expel).
- **Breaking the shift symmetry gets furthest, and fails.** A *global* potential is incompatible as an identity ($dV/dt = V\varphi$ must vanish for $w = -1$, while the condensate needs $\varphi = Q_0 \neq 0$); quantitatively the leak rates lock at $\Gamma n/TV = \rho\Lambda/\rho_{\text{dust}} = 2.60$, so clearing a galaxy in 1 Gyr drifts $\rho\Lambda$ *by 47× and protecting* $\rho\Lambda$ forces a 180 Gyr charge lifetime. A **Y-gated** breaking, $-W(Y/a_0^2)V(\varphi)$ with $W(0) = 0$, *defeats* that — on FRW the symmetry is **exact**, so cosmology is untouched identically. But **unlocking the charge does not unlock the energy:** $\nabla_\mu T^\mu{}_\nu = 0$ holds regardless, and transporting the energy out of a galaxy takes **690 Gyr** at the sector’s own sound speed. Charge decay alone changes nothing about the gravitating mass.
- **** AND LENSING CLOSES IT BY A THEOREM.**** Transforming the equation of state almost works: the dynamical source $\rho + 3p$ vanishes at exactly **$f = 1/3$** , and the field-strength gate makes that a *stable fixed point* rather than a tuning. But **dynamics responds to $\rho + 3p$ while lensing responds to $\rho + p$** — at $f = 1/3$ the lensing source is still $(2/3)\rho$, giving $M_{\text{lens}}/M_{\text{dyn}} \approx 29$ against an observed 1.0–1.3. Generally, $\rho + 3p = 0$ requires $w = -1/3$ and $\rho + p = 0$ requires $w = -1$: **incompatible. No equation of state renders a given energy density invisible to both orbits and light; the only such configuration is $\rho = 0$** , which the transport bound forbids reaching. *The*

no-slip identity $\Phi = \Psi$ that makes this theory's modified-gravity arm work — the property that killed its modified-inertia arm at 21σ and underwrites item 1 — is the same property that forbids hiding the dust.

Status, stated without spin. The galaxy-interior problem is not a missing mechanism: it is a conservation law (energy, via general covariance), plus a transport bound, plus lensing. **The repair space of local, energy-preserving modifications is provably empty.** The defensible slogan is therefore “**no dark-matter particle**” and nothing more; “**none in galaxies**” is **withdrawn as a claim of this theory**. What remains open is a structural change — the Q-sector not acquiring the galactic energy in the first place — against which the smooth-accretion theorem stands for any cold initial condition. Items 1–12 and 15–17 are untouched by all of this.

3. **The post-recombination growth history is NOT verified.** Item 9 covers the acoustic peaks only. Holding c_s^2 at its peak value for all time *destroys* $P(k=0.2)$, so the pass depends essentially on the bump being transient (it peaks at $z \approx 189$). **A patched CLASS fluid carrying $c_s^2(a)$ is required, not optional.**
4. **(v4 rewording — the original referred to withdrawn row 13.)** The dust component's small-scale initial conditions are **not confronted with Lyman- α** . The withdrawn IC route needed a khronon transfer function $T \approx 0.33$ at $k \approx 4.5 \text{ Mpc}^{-1}$ (Λ CDM-excluded); the surviving picture (standard cold dust + the bump response) should be Ly- α -safe by construction, but that is **asserted, not computed — the confrontation remains owed item in the programme.**
5. **Cassini Q_2 is inherited** ($3\text{--}15\sigma$) and is not relieved by anything here.
6. **PPN preferred-frame parameters α_1, α_2** for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.** $\gamma_v = \sqrt{\nu(y_{\text{extN}})}$; the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

5. The falsifiable predictions

prediction	value	how to test
wide-binary boost	$\gamma_v = \sqrt{\nu(y_{\text{extN}})}$	Gaia DR4 , pre-registered and hash-stamped
directional EFE asymmetry	1–4%, signed	already firing at $p = 0.029$ with the right sign

prediction	value	how to test
cluster-to-cluster scatter in the residual	$\sigma(\xi)/\xi \approx 0.02\text{--}0.60$	existing cluster samples — (v4) separates the a_0-bump response (scatter tracks each cluster's distance from the resonance, rows 15–17) from any fixed-scale mechanism, which predicts uniformity (the two prior candidates are withdrawn)
cluster mass profile misfit	ρ_{c} flat, $M_{\text{c}} \sim r^3$, can go negative	fit clusters with the khronon profile instead of NFW
g^{-2} Lorentz violation	computable $s_{\mu\nu}$	SME sector bounds
accelerated structure formation	earlier massive objects	JWST high- z massive galaxies (McGaugh et al. 2024)

6. Credit

$\nu = \sqrt{1+1/y}$ is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złośnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036. $K(Q) = \mu^2(Q-1)^2$ and the ghost- condensate identification: Verwayen, Skordis & Złośnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing bounds that killed this program's first two cluster mechanisms and now two-sidedly bound the third (rows 15–17). GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly- α floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

The distinctive content of this work is the coefficient — $a_0 = \kappa c \sqrt{G\rho_{\Lambda}}$ with $\kappa = \frac{1}{2}$ — its embedding in $_Y$, the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.